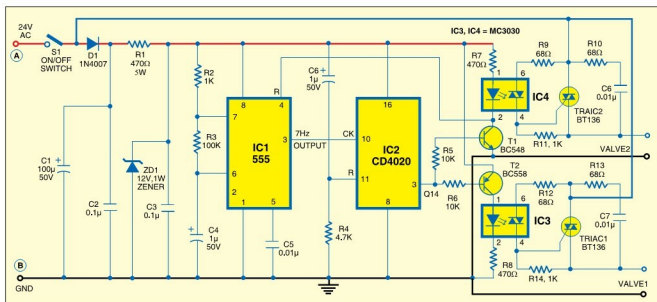




timer circuit.

IC 555 (IC1) is wired as an astable multivibrator producing about 7Hz pulses. You can change this frequency by changing the values of timing components R2, R3 and C4. The output of IC1 is fed to clock pin 10 of IC2. The 7Hz frequency is further divided by IC2, which is a 14-stage binary counter.

IC2 is reset through capacitor C6 and resistor R4 when power switch S1 is closed. Output pin 3 of IC1 goes low when it resets. Counter IC2 is then clocked at 7 Hz, with its final output (Q14) going high after 20 minutes. This high output of IC2 activates the solenoid valves via optocoupled triacs built around IC3 and IC4.

Working of the circuit is simple. When you power-on the circuit using switch S1, IC1 is enabled and IC2 resets. Q14 of IC2 remains low, making transistor T1 cut-off. Transistor T2 conducts to forward-bias the internal

LED of optocoupler IC3. Triac1 is controlled through IC3, which provides gate current through resistor R12 to turn it on. As a result, solenoid valve-1 activates to provide path for water to flow.

After 20 minutes, Q14 of IC2 goes high, making transistor T1 conduct. Transistor T2 cuts off and the internal LED of optocoupler IC4 is forward-biased. Triac2 is controlled through IC4, which provides gate current through resistor R9 to turn it on. As a result, solenoid valve-2 activates to give path for water flow.

Timer IC1 is disabled as soon as the output of IC2 goes high, making its reset pin 4 low to stop clocking to IC2 and so its output remains high. Triac2 has a snubber circuit built around R10 and C6. Triac1 too has a snubber circuit built around R13 and C7. The snubber circuit is used to suppress any voltage spike produced by switching the solenoid valve coils.

Assemble the circuit on a general-purpose PCB and house in a water-proof plastic box with an outlet hole cut in it for the wiring. Seal the wiring hole and box lid with a generous application of silicon sealant.

Before installing the circuit in the sprinkler control system, test it by providing 24V AC as the input and using suitable light bulbs as loads on the output.

Once the timer is ready for installation, remove one control valve from the controller and replace it with 24V AC input of this circuit (terminals 'A' and 'B' in the circuit diagram). Connect the ground terminal of the circuit to the solenoid valve as shown in the circuit.

The timer is set here such that valve 1 remains open for 20 minutes and after 20 minutes, valve 1 closes and valve 2 opens for the following 20 minutes. Valve 2 remains open until switch S1 is closed. ●

NIGHT-VISION DIGITAL GOGGLE

■ UTTAM KUMAR

At night, on highways, the high-intensity light from the headlight of heavy vehicles

blinds oncoming light vehicle drivers (bike riders and car drivers) for a moment. This is the main reason for a large number of accidents on highways at night. By wearing this digital gog-

gle, however, light vehicle drivers can drive the vehicle easily without any

